Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. **True** or **False**: If A is a real matrix and λ is an eigenvalue of A with $\text{Im}(\lambda) \neq 0$ and corresponding eigenvector $\mathbf{u}$, then $\text{Im}(\mathbf{u}) \neq 0$. 1 / 1 point

   - (•) True
   - ( ) False

   Nice work

2. **True** or **False**: If the 2×2 invertible matrix A has hidden rotation–dilation matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where $a, b \neq 0$ then A^{-1} has hidden rotation–dilation matrix $\begin{bmatrix} 1/a & -1/b \\ 1/b & 1/a \end{bmatrix}$. 1 / 1 point

   - ( ) True
   - (•) False

   Nice work

3. **True** or **False**: If A is a real matrix, and λ is a complex eigenvalue of A, then $\overline{\lambda}$ is also an eigenvalue of A. 1 / 1 point

   - (•) True
   - ( ) False

   Nice work

4. **True** or **False**: If λ is an eigenvalue of the real $n \times n$ matrix A with corresponding eigenvector $\mathbf{u}$, then $\text{Re}(\lambda)$ is an eigenvalue of A with corresponding eigenvector $\text{Re}(\mathbf{u})$. 1 / 1 point

   - ( ) True
   - (•) False

   Nice work